

TRANSFORMER & ATTENTION CHEATSHEET

AI Engineer Ready · Week 2 — How Transformers Focus on Context

Golden Rule → Query, Key, Value → score → softmax → weighted sum of Values

1. Transformer Core Blocks

Block	Role
Input embeddings + positional encoding	Turn tokens into vectors and inject word-order information
Multi-head self-attention	Let every token look at every other token and weigh relevance
Feed-forward network	Apply a learned transformation to each token's representation
Residual connections + layer norm	Keep gradients stable so deep stacks can train

A transformer stacks these blocks many times (e.g. 12, 24, 96+ layers) to build increasingly rich representations of the input.

2. What Is Attention? (Intuition)

Example sentence: "The animal didn't cross the street because it was tired." Attention lets the model connect "it" → "animal" using context — every token can attend to every other token in parallel, which is the key shift from older RNN-style models (better long-context handling, faster training at scale).

3. Query / Key / Value

Vector	Question it answers
Query (Q)	"What am I looking for?" — computed for the token doing the attending
Key (K)	"What am I?" — computed for every token being attended to
Value (V)	"What information do I carry?" — what actually gets mixed into the output

4. The Formula

```
$ Attention(Q, K, V) = softmax( Q · K^T / sqrt(d_k) ) · V
$ ^ ^ ^ ^
$ normalize query-key scaling mix in
$ to sum=1 similarity factor values
```

Each output row is a **weighted average of the Value vectors**, where the weights come from how well each token's Key matches the current token's Query. Every attention row's weights sum to 1.

5. Encoder vs Decoder vs Seq2Seq — 30-Second Decision Rule

If the output is...	Use	Example task
A label	Encoder-only (e.g. BERT)	Spam detection: email text → Spam / Not Spam
Open-ended continuation	Decoder-only (e.g. GPT)	AI chat reply: prompt → free-form response

If the output is...	Use	Example task
Input transformed to output	Encoder-Decoder (e.g. T5/BART)	Summarize meeting notes -> short summary

Why it matters

Encoder-only models are built to understand/classify; decoder-only models are built to generate; encoder-decoder models are built to transform one sequence into another (translation, summarization).